

UNAHA, Russian Federation

WMO: 311990

Lat: 55.033N

Lon: 126.800E

Elev: 448

StdP: 96.06

Time Zone: 9.00 (E09)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -43.0	(c) -40.6	(d) -45.4	(e) 0.0	(f) -42.4	(g) -43.0	(h) 0.1	(i) -40.1	(j) 5.5	(k) -21.1	(l) 5.0	(m) -19.5	(n) 0.0	(o) N/A	(p) 0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	27.7	17.2	26.0	16.8	24.2	16.2	19.5	24.4	18.5	23.2	17.5	21.8	1.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.6	21.3	16.9	12.7	20.0	15.8	11.9	19.3	57.7	24.2	54.4	23.1	51.2	21.7	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.3	4.3	3.6	-46.4	31.2	1.5	1.0	-47.5	31.9	-48.4	32.5	-49.3	33.0	-50.4	33.7		
			-46.1	21.2	1.2	1.1	-47.0	22.0	-47.7	22.7	-48.4	23.3	-49.2	24.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.0	-27.6	-23.1	-14.4	-2.4	6.7	14.1	16.4	13.8	6.5	-5.3	-19.1	-26.7
	DBStd	16.64	5.82	5.68	5.32	5.05	4.33	3.49	2.90	2.69	4.07	5.67	5.11	5.67
	HDD10.0	5952	1165	926	756	372	120	6	0	4	119	474	872	1138
	HDD18.3	8532	1423	1159	1014	622	360	132	73	143	354	733	1122	1396
	CDD10.0	482	0	0	0	0	18	129	199	121	15	0	0	0
	CDD18.3	19	0	0	0	0	0	5	13	1	0	0	0	0
	CDH23.3	568	0	0	0	0	35	198	263	71	1	0	0	0
	CDH26.7	102	0	0	0	0	6	35	54	7	0	0	0	0
(14) Wind	WSAvg	1.0	0.8	0.9	1.0	1.3	1.4	1.0	0.8	0.8	0.9	1.1	0.9	0.8
(15) Precipitation	PrecAvg	585	6	7	12	39	50	93	124	111	83	28	18	11
	PrecMax	875	20	21	48	76	151	184	237	260	178	76	51	48
	PrecMin	355	0	0	1	0	12	23	43	24	8	2	3	0
	PrecStd	114	6	5	10	21	33	48	53	54	41	19	11	10
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.7	-5.1	6.1	14.7	27.0	29.5	30.3	27.6	22.7	13.3	-3.0	-9.4
		MCWB	-10.2	-7.5	0.6	6.1	14.4	17.6	18.5	17.9	13.8	6.3	-5.1	-10.8
	2%	DB	-11.8	-7.9	1.7	11.3	23.6	27.5	28.1	25.2	19.8	9.2	-6.0	-12.7
		MCWB	-12.8	-9.5	-2.3	4.2	12.8	16.5	18.0	16.7	12.9	3.3	-7.8	-13.7
	5%	DB	-15.2	-10.3	-0.7	8.7	20.3	25.7	26.4	23.5	17.4	6.2	-8.2	-14.8
		MCWB	-15.9	-11.8	-3.5	3.1	10.7	15.7	18.1	16.2	11.6	1.6	-9.6	-15.6
	10%	DB	-18.1	-12.8	-3.0	6.6	16.9	23.5	24.3	21.5	14.8	3.5	-10.4	-17.3
		MCWB	-18.8	-14.0	-5.4	1.4	8.7	15.2	17.4	15.3	10.4	0.3	-11.6	-18.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.4	-7.3	0.8	6.9	15.1	20.3	21.0	19.9	16.4	6.5	-5.2	-10.7
		MCDB	-9.0	-5.6	5.3	12.2	24.5	25.2	25.6	23.6	19.4	12.4	-3.3	-9.7
	2%	WB	-12.7	-9.4	-1.6	4.7	13.1	18.5	19.9	18.3	14.3	3.7	-7.7	-13.6
		MCDB	-11.6	-7.9	1.0	11.0	22.7	23.5	25.3	22.0	16.7	8.3	-6.2	-12.8
	5%	WB	-16.0	-11.8	-3.5	3.0	11.2	17.3	19.1	17.2	12.9	2.2	-9.6	-15.7
		MCDB	-15.2	-10.3	-0.6	8.2	19.3	22.6	24.1	21.3	15.9	5.6	-8.3	-14.9
	10%	WB	-18.6	-14.0	-5.7	1.7	9.3	16.2	18.2	16.2	11.4	0.6	-11.5	-18.0
		MCDB	-18.0	-12.9	-3.1	5.9	15.9	21.8	23.0	20.1	14.5	3.3	-10.5	-17.4
(35) Mean Daily Temperature Range	5% DB	MDBR	14.4	17.6	18.3	13.5	14.8	15.2	13.2	14.2	13.7	14.1	14.3	13.0
		MCDBR	15.5	18.4	19.8	17.0	22.6	20.0	18.4	18.7	18.8	17.0	16.5	14.0
		MCWBR	14.5	16.4	16.2	10.9	12.0	9.9	9.2	10.8	11.8	11.4	14.6	13.1
	5% WB	MCDBR	15.4	18.0	18.2	15.7	21.4	15.5	14.4	15.3	13.4	16.1	16.4	14.0
		MCWBR	14.4	16.3	14.9	10.2	11.7	8.5	8.2	10.2	9.2	11.0	14.6	13.1
(40) Clear-Sky Solar Irradiance	taub		0.239	0.273	0.317	0.439	0.489	0.474	0.481	0.406	0.353	0.323	0.265	0.234
	taud		2.132	2.105	2.041	1.857	1.926	2.075	2.093	2.309	2.398	2.274	2.121	2.071
	Ebn at Noon		699	800	840	769	757	777	762	807	809	731	640	602
	Edh at Noon		61	91	125	176	175	153	147	112	89	78	61	51
(44) All-Sky Solar Radiation	RadAvg		1.09	2.25	4.07	5.44	5.13	5.76	4.96	4.30	3.19	2.06	1.23	0.78
	RadStd		0.04	0.07	0.14	0.36	0.38	0.45	0.56	0.40	0.29	0.14	0.08	0.03

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page